

Distributed Sterile-Injectable Manufacturing Under Real-World Constraints: An Engineering and Operations Feasibility Study

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2026

STATUS: PRELIMINARY AND ILLUSTRATIVE. NOT FOR EXTERNAL CLAIMS. Every model input except four release-time components is illustrative. Of 91 registered parameters, 87 carry evidence tier 5, the study's own label for a placeholder with no operational, official, peer-reviewed or elicited source behind it; the four exceptions are the sterility incubation, environmental monitoring, assay and endotoxin components of release time. No interview and no expert review has been completed: zero of a protocol target of 25 to 30 interviews, and zero of three required independent reviews. All sixteen regulatory gates are UNCERTAIN with no reviewer assigned, so regulatory eligibility returns NO_CONCLUSION for every architecture in every run quoted here. Nothing in this document is a claim that any system is FDA-compliant, clinically deployable, or ready to manufacture drugs, and no number in it is a statement about sterile-injectable manufacturing in the world. Every figure below is the behaviour of a computational model under those inputs. This paragraph is the standing qualification on every sentence, figure and table that follows.

Abstract

This memo asks under what product, demand, regulatory and operating conditions regional manufacturing of sterile injectables could reduce shortages more effectively than safety stock, dual sourcing, or additional centralized capacity. It compares those options on two non-controlled small-molecule presentations: sodium bicarbonate 8.4 percent in a 50 mL vial, in prolonged shortage, and norepinephrine bitartrate 1 mg/mL in a 4 mL vial, resolved. The method is a protocol frozen before any result, a deterministic screen reconciled cell for cell against the source spreadsheet, a daily discrete-event simulation in which every strategy faces the same simulated world, optimization of each strategy to one pre-specified service target, and an ablation harness over named failure mechanisms. The answer is conditional and largely negative: what decides the outcome is capacity relative to demand and correlated failure across nominally separate sites, and neither is improved by distributing manufacture as such. The study retracted three of its own headline conclusions after correcting defects in its own model, and that history is reported here rather than removed.

1 The problem, the two products, and the question

1.1 What the record supports

United States sterile injectable shortages are counted by two registries that do not count the same object. The ASHP and University of Utah series recorded 323 active shortages in the first quarter of 2024, the highest since tracking began in 2001. The FDA CDER database, which is the statutory list, carried 102 as of 31 July 2024 in the count the Government Accountability Office records. Both are correct on their own definitions

*Prepared with AI coding assistance. What an agent produced, and what the author has and has not reviewed, is recorded in AI_ASSISTANCE.md in the package root.

and the gap is roughly threefold. This package flagged that its own earlier market research quoted the 323 figure without saying it was the ASHP count, which overstates the FDA-defined problem. The distinction is operational, because the FDA list is the legal object that gates the 503B exception, the CMS buffer-stock payment and most contracting language, so a claim to reduce shortage days is measured against that list.

Three features of the record bound what a manufacturing intervention can do. Demand, not plant failure, is the dominant reported driver: increased demand contributed to 66 of the 102 shortages on the FDA list at that date. That driver is also the least observable, because manufacturers must report an interruption caused by increased demand but need not report an increase in demand that does not interrupt production, and FDA has asked Congress for that authority. And concentration risk is not only offshore: the largest recent supply event followed a hurricane closing a single domestic plant in Marion, North Carolina, described by USP as having made roughly 60 percent of US intravenous solution supply.

The prior-art review behind this study covered fourteen solution families over 798 source rows, 651 fetched and quoted, with citation audits on 2 and 3 September 2026. The audits found no fabricated source, company, patent, contract or DOI, and found instead a recurring defect of misattribution, load-bearing numbers cited to sources that do not contain them, with thirty-six corrections in four families alone. The structural finding is that the space is occupied. A centralized plant built specifically for shortage injectables is already implemented rather than proposed, and shortage detection, risk scoring, ingredient mapping, essential-medicines curation and committed-volume contracting each have a funded incumbent with better data than a new entrant would have. What is unsettled is the economics and operations of a distributed alternative, which is why this is a feasibility question rather than a novelty question, and why it is built so the answer may be negative.

1.2 Two products, from a screen that selects nothing

The protocol requires a dated longlist of at least five presentations from frozen FDA and ASHP snapshots including one resolved or recurrent comparator, then hard gates, then weighted scoring of survivors. Six were screened on 2 September 2026 against snapshots whose hashes are recorded in `data/product_dossiers/screen_manifest.json`.

Table 1: Product screen, 2 September 2026, against frozen official snapshots. Shortage rows from the FDA drug shortage export (1,635 rows). No candidate passed every gate, because gates PG5 to PG8 cannot be evaluated from public data.

Presentation	Protocol role	Binding gate result	FDA rows (current)	Outcome
Sodium bicarbonate 8.4%, 50 mL vial	case candidate	PG1 to PG3 PASS	19 (15)	UNCERTAIN
Norepinephrine bitartrate 1 mg/mL, 4 mL	resolved comparator	PG1 to PG3 PASS	0 (0)	UNCERTAIN
Furosemide 10 mg/mL vial	reserve	PG1 UNCERTAIN	33 (30)	UNCERTAIN
Sterile water for injection, vial	scale question	PG1 UNCERTAIN, PG8 mismatch	21 (21)	UNCERTAIN
Acyclovir sodium 50 mg/mL vial	reserve	PG1 UNCERTAIN	0 (0)	UNCERTAIN
Acetazolamide sodium 500 mg vial	out-of-archetype	PG2 FAIL, lyophilized	0 (0)	FAIL

The two carried forward are sodium bicarbonate injection 8.4 percent in a 50 mL single-dose vial and norepinephrine bitartrate 1 mg/mL in a 4 mL single-dose vial, chosen for the contrast the protocol asks for. Sodium bicarbonate is the prolonged current shortage: 19 rows in the FDA export, 11 matching the 8.4 percent presentation, 15 current, earliest posting 1 March 2017, stated reasons demand increase, discontinuation and other. Norepinephrine is the resolved comparator: zero rows in the FDA export and the openFDA mirror on the same date, and one ASHP bulletin ever, opened 9 February 2017 and resolved 7 June 2022. Both are non-controlled small-molecule aqueous solutions in standard vials, the archetype the modelled process train represents, and both clear the identity, manufacturing-fit and control-status gates that acetazolamide fails on lyophilization. Their supplier structures differ, 13 approved holders against 23, and under the illustrative inputs one network is capacity-short by construction and the other is not. That contrast is the point of the pair: a conclusion holding on only one of them is a conclusion about a product rather than about an architecture.

Four qualifications attach. The screen selects nothing, because gates PG5 to PG8, covering demand observability, clinical substitutability, input feasibility and economic relevance, cannot be evaluated from public data, so every survivor ends UNCERTAIN and the protocol's rule is that UNCERTAIN blocks final selection. The protocol's step 5 requires a hospital pharmacy leader and a sterile-manufacturing expert to challenge the selection independently, and that has not happened. The selection record contains zero finalized presentations and both product configurations carry a header saying they are provisional. And the source workbook labelled norepinephrine "resolved/recurrent" where the snapshot does not support the recurrent half. These are working presentations for a model, used as such.

1.3 The question

Under what product, demand, regulatory and operating conditions could regional manufacturing reduce sterile-injectable shortages more effectively than safety stock, dual sourcing, or additional centralized capacity?

The words that carry weight are "under what conditions". The question is not whether distributed manufacturing is a good idea but which region of the product, demand, regulatory and operating space, if any, contains a distributed design that beats the alternatives on cost at a matched service level. A negative answer, or one holding only inside a narrow and clearly stated region, is a legitimate result and the study was built to permit one. The comparison set is fixed in the protocol as the eight strategies S0 to S7 of Table 6, which implement the four comparator classes the specification names and add centralized expansion and a 503B responder; twelve further architectures were configured later as S8 to S19 and are compared on the same terms. Quality and regulatory requirements enter as hard constraints throughout. They are never relaxed to improve an economic result, and an ineligible strategy is excluded rather than charged a penalty an optimizer could pay through.

2 Method

The protocol was written and frozen first, as `protocol/protocol.yaml` version 1.0.0, converted on 1 September 2026 from the author's protocol document and the source workbook. The service target was fixed before any simulation result existed: mean fill rate at least 0.99, and probability at least 0.90 that a run's own fill rate reaches 0.99, with a shortage day defined as a day whose fulfilled fraction falls below 0.95. Those values are recorded in revision R001 on the day of the freeze, with the note that no simulation result existed at that date. Eleven revisions have been logged since, ten flagged decision-relevant, each carrying a date, section, change, rationale, author, approval state and affected files (Table 2). Most are corrections the study made to its own model after finding a defect in it. None has been signed off, so the definition-of-finished gate for protocol freeze fails and is recorded as failing.

The study inherited an Excel model and reproduces it rather than replacing it silently. `deterministic.run_screen` in replica mode recomputes the screen for all sixteen rows from the same base inputs, and `reconcile_with_workbook` reports any absolute difference above 10^{-6} for currency or 10^{-9} for fractions. All 448 cached values reconcile, the workbook snapshot is pinned by SHA-256 hash, and the reconciliation is a regression test rather than a one-time exercise. A separate corrected mode applies the defects the workbook audit found, including a delivered-unit denominator and a capacity cap, and reports its deltas separately, so a correction cannot be misread as a reconciliation failure. The screen is a verification device and produces no service verdict, because a deterministic screen cannot produce one.

Service and cost come from a daily discrete-event simulation of sites, suppliers, regions and lanes. Each run generates one exogenous world for a universal roster of entities, and every strategy in that run consumes the subset it instantiates, so paired strategies see identical demand, shocks, site failures, supplier disruptions, common-cause events, shortage-list transitions and transit noise. This is the study's first fair-comparison rule and it is enforced by test. Random draws are keyed by run index, named stream, a BLAKE2b hash of the

Table 2: The eleven logged protocol revisions (`protocol/revisions.csv`). “DR” marks one flagged decision-relevant. None has been signed off.

Id	Date	Subject	DR
R001	2026-09-01	Service thresholds pre-specified before any result	yes
R002	2026-09-01	Regulatory gate set adopted, every status UNCERTAIN	yes
R003	2026-09-02	Raw-material stock carries the inventory carrying rate	yes
R004	2026-09-03	Expansion lead time, cost accrual, opening cohorts, reserved lines	yes
R005	2026-09-03	Reorder point, node placement, utilization denominator, error bars	no
R006	2026-09-03	Twelve further architectures configured as S8 to S19, gate-mapped	yes
R007	2026-09-03	Supplier capacity as a fraction, reserved capacity screened, gate G16	yes
R008	2026-09-03	Commissioning clock starts at the first measured day	yes
R009	2026-09-05	Opening inventory is purchased and charged, not inherited free	yes
R010	2026-09-05	Every comparator searches one common inventory space	yes
R011	2026-09-05	Pipeline cover, region weights, readiness clock, contract priority	yes

entity identifier and an ordinal, so adding an entity to a design never perturbs another entity’s draws, which is what makes a comparison between two designs a comparison of the designs.

No strategy is compared at its default settings. For each product and strategy the optimizer reads regulatory eligibility from the gates first, so an excluded strategy is never optimized and a strategy whose gates are unresolved is optimized for information and labelled as such. It searches the strategy’s declared design space on a common random-number batch, where feasible means the mean fill rate reaches the target and the share of runs reaching it clears the tail confidence, refines the best feasible point by coordinate search, and re-evaluates the incumbent at full sample size with its Monte Carlo standard error. If no point is feasible the record says so and reports the closest fill rate. Nothing pays a penalty to become feasible.

Attribution comes from an ablation harness of 22 named failure mechanisms, each a set of parameter overrides applied through the same provenance-carrying records the rest of the study uses, so a switched-off mechanism is a stated value visible in the run manifest. For each product and strategy the harness runs a base configuration, a leave-one-out configuration per mechanism, an all-off configuration, and an add-one-in configuration per mechanism. Leave-one-out and add-one-in are the two ends of the Shapley averaging order, so the harness reports the bracket between them rather than a single attributed share, and reports the Monte Carlo standard error of every mean, on the rule that a difference smaller than the standard errors it is built from is not evidence of anything.

The harness turned out to be a defect detector as much as an attribution tool, because a mechanism whose measured sign is wrong is usually a bug. That is how most of the twenty-four model defects in `bottleneck_decomposition.md` section 7 were found, and it is why this memo carries a correction history. After defects NEW-1, MD-3, MD-11 and MD-12 were fixed under revisions R009 to R011 and every incumbent design was re-evaluated at full resolution without re-searching, three statements the study had presented as headline conclusions were falsified by its own corrected model and withdrawn; Section 7 reports them. Three limits bound everything that follows, beyond the standing qualification on page 1. The ablation measures only mechanisms the engine represents, so the absence of a mechanism is not evidence that it does not matter; known omissions include multi-product pooling, campaign scheduling across products, and a second inventory stage for postponement. The engine models one product per network over four equal regions. And the entire input set is illustrative, so the ranking of mechanisms, like every cost and service number here, is a property of those values. Where a result would change under a different input, the memo reports the condition rather than the number.

3 The engineering, operations and reliability model

The model has two layers. A deterministic screen implements the protocol’s closed-form equations and reconciles them against the source workbook. A stochastic daily network simulation over five measured

years after a 365-day warm-up produces every service and cost number reported. The screen audits arithmetic and bounds and never ranks architectures, because its shortage-day column ignores safety stock and release queues. Apart from the four release components marked tier 1 and 4 in Table 3, every input below is tier 5, illustrative.

3.1 Capacity, unit cost, response time, break-even

Saleable annual capacity over n sites with q units per batch, b nominal batches per site-year, yield y and uptime u , and capacity utilization against annual demand D , are

$$C = n q b y u, \quad U = D/C. \quad (1)$$

Yield and uptime enter the model only here, so a proportional change in either is arithmetically the same lever as the same proportional change in b . Capital is annualized by the capital recovery factor at discount rate r over economic life L , with the limit $1/L$ at $r = 0$; fully loaded cost per delivered unit is the sum of the ten annual ledgers divided by delivered non-expired units Q :

$$\text{CRF}(r, L) = \frac{r(1+r)^L}{(1+r)^L - 1}, \quad c = \frac{1}{Q} (K + F + V + M + O_{\text{inv}} + T + W + I + R + G), \quad (2)$$

where K is annualized capital, F fixed site operations including the quality unit and labour, V product and site launch, M variable production, O_{inv} the opening inventory purchase, T quality testing, W failure, scrap and expiry waste, I inventory carrying and distribution, R resilience contracts and G operating-system integration. The denominator is delivered non-expired units, not demand: the workbook divided by demand, which credits a plant for units it cannot make. Break-even committed volume, the resilience premium over committed units, and incremental cost per shortage-day avoided against a reference are

$$Q_{\text{be}} = \frac{F + R}{m}, \quad p = \frac{C_s - C_0}{Q_{\text{committed}}}, \quad \text{ICER} = \frac{C_s - C_{\text{ref}}}{SD_{\text{ref}} - SD_s}, \quad (3)$$

with m the contribution margin per delivered unit, price net of variable production, testing, distribution and expected waste, and SD shortage days. The ratio is computed only where the comparison is a genuine trade-off; a strategy that costs more and avoids no shortage days is labelled dominated and receives none. Price, reimbursement, willingness to pay, provider shortage cost and production cost are separate arguments in the economics module and are never summed. CMS average sales price and Part B spending appear nowhere as a manufacturing cost.

Response time as the workbook sums it is serial with no overlap, $T_{\text{resp}} = \ell_{\text{mat}} + t_{\text{change}} + t_{\text{cycle}} + \phi t_{\text{rel}} + t_{\text{deliv}}$, over raw-material lead time, changeover, production cycle, release time scaled by a strategy factor ϕ , and delivery. The protocol warns that these overlap only where overlap is operationally valid, so the stochastic engine does not use that sum. It runs setup, fill, test and release as queues and reports time to first response and time to stable recovery, the second being the first day followed by 30 consecutive days at or above the daily shortage threshold of 0.95.

3.2 The daily engine and the release path

Each simulated day runs the protocol's nine-step order: expire lots first-expiry-first-out, advance demand, advance disruption states, receive materials and released batches, serve regional demand within a backorder window, start batches where the finished-goods position falls below target and materials permit, replenish and trigger contract or transfer policies, accrue the ten ledgers, then assert a mass balance in integer units,

$$I_0 + P_{\text{saleable}} = H + T_{\text{transit}} + Q_{\text{release}} + E + S, \quad (4)$$

opening stock plus saleable production equals on hand plus in transit plus in the release queue plus expired plus served. The assertion caught a real leak on the first full-horizon run, a shipment that expired in transit and was counted as served. Line occupancy is $365.25/b/u$ days per batch, stretched by the reciprocal of the site’s available capacity fraction that day and floored at changeover plus production cycle, so uptime is a time stretch rather than a discount on output.

Release time is not one number. Sterility incubation, environmental monitoring, assay and endotoxin start together when the batch is filled, so the critical path is their maximum, and quality-unit review is serial after them:

$$t_{\text{rel}} = \max\{t_{\text{ster}}, t_{\text{EM}}, t_{\text{assay}}, t_{\text{endo}}\} + t_{\text{QA}}. \quad (5)$$

At base values this is $\max\{14, 7, 5, 2\} + 2 = 16$ days, of which 14 are sterility incubation and are not reducible by any analytical method the study models. Release scenario R1, which reduces administrative time, can act only on the serial term, and R3, which could remove the assay component, sits off the critical path unless sterility is shortened first. R3 cannot be instantiated while gate G15 is UNCERTAIN. One defect is stated rather than hidden: the release-time factor ϕ is read only by the deterministic screen and never by the stochastic engine (MD-18), which is one of two reasons architecture S6 returns service numbers identical to S5. The aggregate release inputs of 14 and 12 days in Table 4 and the 16-day decomposition disagree; that is a known inconsistency between screen and engine, and the engine uses Eq. 5.

Table 3: Release-time components, from `config/global.yaml1`. The first four are the only inputs in the model above tier 5.

Component (parameter id)	Path	Base (d)	Range (d)	Tier
sterility_incubation_days	parallel	14	14 to 18	1
environmental_monitoring_days	parallel	7	5 to 7	4
assay_days	parallel	5	1 to 7	4
endotoxin_days	parallel	2	0 to 3	4
qa_review_days	serial	2	0 to 4	5
release_time_cv	dispersion	0.20	0.10 to 0.40	5
investigation_duration_days	added on deviation	21	7 to 60	5

3.3 Operational and cost parameters

Table 4 holds the operational inputs the specification names, all tier 5. Node count is not a single parameter: the network is four equal regions by protocol choice, with 3 and 8 as sensitivity settings, and site count is a design variable the optimizer searches. Nodes may exceed the region count and are placed round-robin, and a second node inside one region shares that region’s geography common-cause group, so redundancy inside a region buys less than redundancy across regions.

Cost inputs are tier 5 on the same low, base, high convention. Per unit: materials 0.60 USD for sodium bicarbonate and 0.55 for norepinephrine (0.25 to 1.50 and 0.20 to 1.50), conversion 0.60 for both (0.20 to 1.50), distribution 0.20 (0.05 to 0.80). Per batch: testing 30,000 USD (10,000 to 100,000). Per site: fixed quality and labour 3.0 million USD per year (1.5 to 6.0), installed capital 40 million USD (15 to 100), validation and transfer 4 million USD once (1 to 12). Globally: discount rate 0.10 (0.06 to 0.15) over a 10-year life (7 to 15), inventory carrying 0.20 per year (0.12 to 0.30), and a shared operating layer at 0.30 million USD per site-year (0.10 to 0.80) charged only outside release scenario R0. The last is an open defect (MD-25), because every design-space architecture runs R0 and so declares a shared software group carrying hazard with no cost line.

3.4 Reliability

Reliability is where the architectures separate, so failure processes are modelled explicitly rather than assumed into a service level. Five mechanisms generate the exogenous world, all at the tier-5 rates of Table 5. Daily regional demand is annual demand times regional share divided by 365.25, times growth, seasonality,

Table 4: Operational parameters, base value with declared low and high range. All tier 5, illustrative, from the two product configurations and `config/global.yaml`.

Parameter id	Units	Sodium bicarbonate	Norepinephrine	Tier
annual_demand_units	million units/yr	1.20 [0.60, 2.40]	0.90 [0.50, 1.80]	5
units_per_batch	thousand units	25 [10, 60]	30 [10, 75]	5
batches_per_site_year_nominal	batches/site-yr	45 [30, 70]	45 [30, 70]	5
yield_fraction	fraction	0.96 [0.90, 0.99]	0.96 [0.90, 0.99]	5
uptime_fraction	fraction	0.85 [0.70, 0.95]	0.85 [0.70, 0.95]	5
production_cycle_days	days/batch	3 [1, 7]	3 [1, 7]	5
changeover_days	days	3 [1, 10]	3 [1, 10]	5
release_time_days	days/batch	14 [5, 28]	12 [5, 28]	5
material_lead_time_days	days	90 [30, 240]	90 [30, 240]	5
shelf_life_months	months	24 [12, 36]	24 [12, 30]	5
expiry_scrap_fraction	fraction	0.020 [0.005, 0.080]	0.020 [0.005, 0.080]	5
delivery_days	days	2 [1, 5]	2 [1, 5]	5
target_safety_stock_days	days	60 [30, 180]	60 [30, 180]	5
yield_sd	fraction	0.02 [0.01, 0.05]	0.02 [0.01, 0.05]	5
node_commissioning_days	days	730 [365, 1095]	730 [365, 1095]	5
capacity_expansion_days	days	270 [90, 730]	270 [90, 730]	5

a routine multiplier and a shock multiplier, rounded to integer units; the routine multiplier is a mean-one log-normal AR(1) process at `demand_cv` with lag-1 autocorrelation, so a bad week is correlated with the week before it, and shocks arrive as a Poisson process hitting all four regions with a declared probability and otherwise one, with lognormal magnitude and duration. Sites and suppliers each draw Poisson failure arrivals with lognormal durations; degradation is proportional rather than binary, so a line at capacity fraction f takes $1/f$ times as long and a supplier at f ships at most f of one full order per day. Since revision R011 pipeline cover is separated from the policy cover of safety plus cycle stock, so shortening a lead time no longer shrinks the buffer, though a residual remains and is not argued away, because under a forced outage a deferred order lands when the supplier resumes whatever its nominal lead. A site that does not exist yet contributes nothing but sunk capital, and since revision R008 starts its commissioning clock at the first measured day rather than at simulation day zero; before that fix, three commissioning parameters coincided with the warm-up, so any lead time of a year or less cost nothing inside the measured window, and the architecture that had benefited most from the artifact was dual sourcing.

Nominal site count is not independent redundancy, and the protocol’s fair-comparison rules make that a hard requirement. Every site and supplier declares its common-cause groups, and a site’s available capacity on a day is the minimum of its own and that of every group it belongs to. Groups arrive at `common_cause_events_per_year` with lognormal durations, and impact is a Beta draw, so an event degrades rather than switches off. Six kinds of group are represented: API supplier, container and closure supplier, stopper and seal supplier, shared software or model version, ownership and the quality unit that performs disposition, and geography. The last two carry the study’s scepticism about nominal redundancy. A shared quality unit means one legal holder disposing batches from several sites, which is the situation for most contract-manufacturing architectures whatever their site count; five design-space architectures were configured claiming an independence their own regulatory-owner field denied, and revision R007 added the shared quality group to all five. The same revision fixed a defect that had made supplier-level groups structurally inert, since shipment was gated on capacity strictly above zero while a group only ever sets capacity to $1 - \text{impact}$, so a shared key-starting-material tier could never delay a unit at any parameter value (MD-19). Fixing the structure did not make the mechanism bite at the frozen hazard rate, which is reported as a null rather than dropped.

3.5 The regulatory state machine

The sixteen gates are binary feasibility constraints, not costs. A FAIL excludes an architecture. UNCERTAIN permits simulation for information only and can never receive a favourable decision class, and there is no

Table 5: Reliability parameters, base with declared range, all tier 5 illustrative, from `config/global.yaml`. Durations are lognormal medians at dispersion `disruption_duration_cv`.

Mechanism	Parameter id	Base	Range
Routine demand	<code>demand_cv</code>	0.15	0.08 to 0.30
	<code>demand_autocorrelation</code>	0.30	0.00 to 0.70
	<code>annual_demand_growth</code>	0.01	−0.02 to 0.05
Demand shocks	<code>demand_shocks_per_year</code>	0.40	0.10 to 1.00
	<code>demand_shock_multiplier</code>	1.35	1.10 to 2.00
	<code>demand_shock_duration_days</code>	30	7 to 90
Site failures	<code>demand_shock_all_regions_probability</code>	0.50	0.20 to 0.90
	<code>site_failures_per_site_year</code>	0.20	0.05 to 0.60
	<code>site_failure_duration_days</code>	45	7 to 180
	<code>site_failure_residual_capacity</code>	0.00	0.00 to 0.50
Supplier disruptions	<code>supplier_disruptions_per_supplier_year</code>	0.20	0.05 to 0.60
	<code>supplier_disruption_duration_days</code>	45	14 to 120
Common cause	<code>common_cause_events_per_year</code>	0.10	0.02 to 0.30
	<code>common_cause_duration_days</code>	60	14 to 240
	<code>common_cause_capacity_impact</code>	0.50	0.20 to 1.00
	<code>common_cause_capacity_impact_sd</code>	0.15	0.05 to 0.30
Quality	<code>deviation_rate_per_batch</code>	0.020	0.005 to 0.080
	<code>batch_rejection_rate</code>	0.010	0.002 to 0.040
Regulatory state	<code>shortage_list_resolution_rate_per_year</code>	0.30	0.10 to 1.00
	<code>shortage_list_relisting_rate_per_year</code>	0.10	0.02 to 0.40
	<code>bud_503b_days</code>	90	45 to 180
Logistics	<code>transport_time_cv</code>	0.25	0.10 to 0.50
	<code>disruption_duration_cv</code>	0.60	0.30 to 1.00

penalty path an optimizer can pay through. As of the current runs all sixteen are UNCERTAIN with no reviewer assigned, so eligibility is NO_CONCLUSION for every one of the twenty architectures. The 503B pathway is not a permanent pathway and is not modelled as one. It is a time-varying legal state,

$$\text{eligible}(t) = \text{listed}(t) \wedge \text{gates}_{\text{static}} \wedge \text{permission}(\text{region}), \quad (6)$$

where $\text{listed}(t)$ is a two-state Markov chain on the FDA shortage list with hazards `shortage_list_resolution_rate_per_year` and `shortage_list_relisting_rate_per_year`, initialised from the frozen 2026-09-02 snapshot; $\text{gates}_{\text{static}}$ requires G08 and G11; and permission requires G12, with regions lacking state licensure removed from the service set. The predicate is checked when a batch is compounded and again on each shipment and issue day, because the statute requires the condition at compounding, at distribution and at dispensing. Section 6 states what that implies.

3.6 The architectures, and what the model does not represent

Table 6 lists the twenty architectures. All are optimized to the same target, minimizing expected annual cost subject to mean fill at or above 0.99 and a probability of at least 0.90 that a run meets it, on common random numbers. Since revision R010 every frozen comparator searches the same inventory space the design-space strategies declared for themselves, because the earlier comparison measured search freedom as much as architecture: added central capacity moved from infeasible to feasible on both products at roughly 60 percent of its declared-space cost. The symmetric test, narrowing the design-space strategies to the comparators’ old space, has not been run and would be the honest completion of the check.

The limits are recorded rather than described around. One product per network, so no campaign scheduling or cost pooling across a portfolio; this is the largest unquantified distortion in the cost comparison, because a node’s fixed cost is borne by a single presentation at low utilization (MD-15). Four equal regions, so three of the four allocation policies are arithmetically the same function at the base case. No supplier allocation rule during scarcity. Provider harm and clinical substitution are not monetized. And the screen’s common-impact factor is derived from topology for S8 to S19 but hand-set for S0 to S7, so no screen-level common-impact comparison across the two arms is valid (MD-22).

Table 6: The twenty architectures. S0 to S7 are the frozen comparators, carrying the four comparator classes the specification names plus centralized expansion and the 503B responder. S8 to S19 are the design-space additions of revision R006.

Id	Architecture	Id	Architecture
S0	Status quo centralized supply	S10	Split tenancy, two independent multi-product hosts
S1	Additional safety stock	S11	Bright-stock campaign offtake, national reserve
S2	Dual source plus contracts	S12	Contracted registered-capacity supply base
S3	Reserved flexible contract capacity	S13	Contracted base, rotated surge, reserve stock
S4	Additional centralized capacity	S14	Cooperative-financed registered second source
S5	Distributed regional nodes, conventional release	S15	Capacity-adequate presentation, no new plant
S6	Distributed nodes plus validated release assurance	S16	Reserve-triggered campaign network
S7	503B shortage response (time-varying legal state)	S17	Rotating prequalified campaigns, three registered lines
S8	Acquired registered line, split API and geography	S18	Dual committed supply, contract-mandated sources
S9	Dual finished-dose source, key-starting-material tier	S19	Maintained alternate-source switching package

4 Results: what the engine returns

Every number here is engine behaviour under evidence tier 5 inputs. All sixteen regulatory gates carry status UNCERTAIN with no reviewer, every run returns eligibility NO_CONCLUSION for every strategy, and nothing below measures any plant, product or supply chain. The frozen requirement is a tail statement, not a mean: a design meets it when $P(\text{run-level fill rate} \geq 0.99) \geq 0.90$ over a five-year measured window after a 365-day warm-up. Both numbers were pre-specified on 2026-09-01 as revision R001, before any simulation result existed. Among designs that meet it, lower annual total cost is preferred. Cost never buys a relaxation of the requirement. Because the requirement is a Monte Carlo proportion, the binomial standard error at the requirement governs what a verdict is worth: the optimization and the ablations run at 60 paired runs (0.039), the dominance and reversal maps at 40 (0.047), the threshold bisections at 20 (0.067), the simulation and the re-evaluation at 100 (0.030). All postdate revisions R009 to R011, so earlier runs are superseded and none appears here.

One structural fact separates the two products in almost every table below. At the illustrative inputs the norepinephrine network’s status-quo saleable capacity runs at 0.778 of demand-implied throughput and the sodium bicarbonate network’s at 1.245. The bicarbonate case is capacity-short by construction, and a network whose saleable capacity is below demand cannot be stocked out of the deficit over a long enough window. That is a property of the illustrative demand and capacity numbers, not an observation about the products.

Four things follow from Table 7, each a statement about the model. First, on norepinephrine the status quo at its own optimized inventory policy already returns mean fill 0.9949 and $P(\text{meet}) = 0.917$ at 13.82 USD per unit, where the configured default design in the 100-run simulation returns 0.9474 and 0.100 at 14.28 USD per unit. The two come from runs at different resolutions, so only the direction should be read, but the direction is large: on the product whose plant is not capacity-short, most of the achievable service comes from inventory policy at the incumbent plant, at roughly 0.2 million USD per year, with no new plant, no second source and no new site. Second, once that baseline exists, further service is expensive: the cheapest architecture that avoids any shortage days on norepinephrine does so at 497 thousand USD per shortage-day avoided per year, and distributed regional nodes at 3.27 million, where on sodium bicarbonate, whose baseline loses 64 shortage days a year, the same two figures are 43 thousand and 306 thousand. The specification’s headline efficiency output differs by nearly two orders of magnitude between the products, and the difference is whether the incumbent network has enough capacity. Third, the release-assurance

Table 7: Specification outputs at each architecture’s optimized design, 60 paired runs, opt_20260906T043357Z. Days avoided is the paired reduction in shortage days per year against S0 on the same seeds; incremental cost is thousands of USD per shortage-day avoided per year and reads “none” where no days are avoided. Response times are medians across episodes. Six architectures are omitted for space and appear in Table 8 where they qualify. All inputs tier 5 illustrative.

Id	USD/ unit	Annual USD M	Mean fill	P(meet dem.)	Short. d/yr	Days avoid.	Incr. kUSD/d	First resp. d	Stable recov. d
<i>Norepinephrine bitartrate 1 mg/mL, 4 mL vial (resolved comparator)</i>									
S0	13.82	12.91	0.9949	0.917	7.5	+0.0	n/a	3.7	4.0
S1	13.82	12.91	0.9949	0.917	7.5	+0.0	none	3.7	4.0
S2	19.22	18.03	0.9986	0.950	1.5	+6.0	848	1.1	8.4
S3	18.70	17.48	0.9952	0.900	9.4	−1.9	none	1.0	2.5
S4	21.53	20.18	0.9975	0.950	2.1	+5.4	1347	0.7	10.3
S5	21.39	20.04	0.9975	0.950	5.3	+2.2	3266	8.3	12.8
S6	22.45	21.03	0.9975	0.950	5.3	+2.2	3721	8.3	12.8
S7	16.93	15.83	0.9956	0.933	6.6	+1.0	3021	6.7	4.3
S9	24.85	23.28	0.9972	0.917	2.0	+5.6	1864	2.6	12.7
S11	16.22	15.15	0.9949	0.933	3.9	+3.7	611	2.8	19.3
S15	13.76	12.87	0.9958	0.917	9.1	−1.6	none	2.1	3.5
S16	17.07	16.00	0.9981	0.950	1.3	+6.2	497	1.9	6.4
S17	19.64	18.38	0.9967	0.950	10.7	−3.1	none	6.0	7.5
S19	17.02	15.88	0.9939	0.917	3.8	+3.7	806	2.2	18.5
<i>Sodium bicarbonate 8.4%, 50 mL vial (prolonged current shortage)</i>									
S0	11.69	13.41	0.9200	0.017	64.2	+0.0	n/a	14.8	0.0
S1	11.69	13.41	0.9200	0.017	64.2	+0.0	none	14.8	0.0
S2	19.26	24.00	0.9949	0.850	15.3	+48.9	217	3.9	19.6
S3	14.82	18.49	0.9963	0.917	3.5	+60.7	84	1.3	10.7
S4	19.23	24.04	0.9984	0.950	1.2	+63.0	169	0.9	7.6
S5	25.63	31.98	0.9961	0.900	3.6	+60.6	306	3.8	13.2
S6	27.02	33.70	0.9961	0.900	3.6	+60.6	335	3.8	13.2
S7	13.77	16.56	0.9632	0.317	32.8	+31.4	100	8.0	0.3
S9	19.68	24.63	0.9993	0.967	1.0	+63.2	177	0.8	6.0
S11	12.89	16.11	0.9982	0.950	1.9	+62.3	43	0.3	4.8
S15	11.77	13.38	0.9122	0.017	69.3	−5.1	none	14.9	0.0
S16	13.40	16.72	0.9961	0.917	3.8	+60.4	55	1.4	24.2
S17	15.57	19.41	0.9958	0.917	15.4	+48.8	123	4.4	8.4
S19	14.77	16.35	0.8881	0.000	99.9	−35.7	none	13.3	271.9

variant costs money and returns nothing measurable: S6 matches S5 to four decimal places on both products while costing 0.99 and 1.73 million USD a year more, and part of that identity is defect MD-18, so the cost difference is real inside the ledger while the service equality is partly an artifact. Fourth, response time is not the binding output: medians to first response run from 0.3 to 14.9 days and to stable recovery from 0.0 to 271.9, and recovery is slowest where fill is already high, because an architecture that rarely stocks out records few episodes and clears each slowly relative to its own small deficit.

Table 8 gives the fourteen of forty product-strategy cells meeting the requirement when each cell’s own incumbent design is re-evaluated at 100 paired runs without re-searching, which separates a change of classification caused by a model fix from one caused by a small final-run count. Eight of the fourteen are borderline, and eight further cells the run calls infeasible also sit within one standard error, sixteen of forty in total. What that means is narrow: at 100 paired runs a cell estimated at 0.900 and one at 0.870 cannot be told apart, so their opposite labels carry no information, and only six cells across both products are separated from the requirement by more than one standard error. Two architectures meet it on both products, rotating prequalified campaigns (S17) and dual finished-dose sourcing with a key-starting-material tier (S9), and S17 returns the only tail probability of 1.000 in the set. The table is a floor rather than a boundary, because it scores incumbents of an earlier and smaller search. The later search over the same space finds a feasible design for fifteen of the twenty-six cells this re-evaluation calls infeasible, including sodium bicarbonate S11 at 0.950 against 0.630 and norepinephrine S0 itself. The disagreement runs one way only, which is the signature of a search that has not converged rather than of a model that has changed. The honest reading is that this package

Table 8: The verified feasible set at 100 paired runs, from `results/design_space/reeval_post_R009.json`. A cell is borderline when its estimated $P(\text{meet})$ lies within one binomial standard error (0.030) of the 0.90 requirement.

Product	Id	Annual USD M	Mean fill	$P(\text{meet})$	SE	Borderline
Norepinephrine	S11	15.2	0.9952	0.900	0.030	yes
Norepinephrine	S19	15.9	0.9943	0.920	0.027	yes
Norepinephrine	S2	18.0	0.9989	0.960	0.020	no
Norepinephrine	S17	18.4	0.9975	1.000	0.000	no
Norepinephrine	S13	18.8	0.9964	0.900	0.030	yes
Norepinephrine	S5	20.0	0.9963	0.940	0.024	no
Norepinephrine	S6	21.0	0.9963	0.940	0.024	no
Norepinephrine	S8	22.5	0.9964	0.910	0.029	yes
Norepinephrine	S9	23.3	0.9975	0.920	0.027	yes
Norepinephrine	S14	23.9	0.9964	0.920	0.027	yes
Bicarbonate	S17	19.4	0.9960	0.900	0.030	yes
Bicarbonate	S12	21.5	0.9967	0.920	0.027	yes
Bicarbonate	S18	24.2	0.9977	0.930	0.026	no
Bicarbonate	S9	24.6	0.9987	0.940	0.024	no

can identify architectures that plainly fail and cannot yet rank the ones that do not.

No price is assumed anywhere. The break-even price is derived from each run’s own cost ledger and delivered volume, and the break-even volume is the committed volume covering fixed and committed cost at that price; no reimbursement figure enters the cost side or stands in for a price. Across all forty cells the fixed share of annual cost runs from 0.797 to 0.937, so every architecture here is a fixed-cost business in the model and its break-even volume is a large fraction of its own capacity: 0.64 to 0.94 overall, and 0.80 or above for every architecture meeting the requirement on either product. On norepinephrine the break-even price is 14.26 USD per unit for the status quo, 16.63 for bright-stock campaign offtake, 20.55 for rotating campaigns, 46.65 for distributed regional nodes and 49.10 once the release layer is added; on sodium bicarbonate the same five are 14.44, 13.28, 15.72, 40.59 and 42.67. These are model cost recoveries under illustrative inputs, not prices, not offers, and not evidence that any buyer would pay them.

5 Sensitivity: what determines the answer

Three mechanisms decide whether an architecture in this battery meets the requirement, and they are not the three the original thesis rests on. In order of measured effect they are capacity adequacy, the regional inventory review rule, and commissioning time, and capacity adequacy dominates the other two by roughly an order of magnitude on every service metric. Release time, the mechanism a release-assurance layer would address, is worth two of the sixteen days in the release path and moves mean fill by less than 0.002 anywhere in the battery. Capital cost and fixed quality and labour cost change feasibility in none of the forty cells at any point of their declared ranges, and in the reversal map they do not change even the ranking.

Two rows of Table 9 carry the result. Annual demand changes feasibility in all forty cells and batches per site-year in thirty-nine of forty; both are statements about capacity adequacy, the first varying the demand a fixed network must serve and the second how much it can make. No other input changes feasibility in even four fifths of the cells, and capital per node and fixed quality and labour per site-year change it in zero cells across a four-fold and a six-and-a-half-fold range, so cost is not a feasibility lever in this model. Two cautions belong with the census: the bisections run at 20 paired runs, where the standard error is 0.067, so a threshold is an interval and not a point; and twelve cells sit in the last column for most inputs and can never show a threshold, so the informative denominator for the weaker inputs is 28 rather than 40. As multiples of the base demand each cell was designed for, the demand thresholds run from 0.71 to 1.17, median 1.075 on norepinephrine and 0.99 on sodium bicarbonate. No architecture buys more than about seventeen per cent of demand headroom over its own design point, and the largest headroom in the battery is the status quo’s

Table 9: One-way threshold census over 40 product-strategy cells per input, 480 bisections at 20 paired runs each, from `ds_post_R009/thresholds.csv`. The last two columns count cells whose classification never changes anywhere in the declared range.

Input	Declared range	Thresh. inside	Feasible throughout	Infeas. throughout
Annual demand	0.5M to 1.8M; 0.6M to 2.4M	40	0	0
Batches per site-year	30 to 70	39	1	0
Common-cause events per year	0.02 to 0.3	31	4	5
Changeover days	1 to 10	25	3	12
Shelf life, months	12 to 30; 12 to 36	19	9	12
Raw-material lead time, days	30 to 240	19	11	10
Supplier concentration	0.05 to 0.6	17	14	9
Release time, sterility, days	14 to 18	6	22	12
Node commissioning, days	365 to 1095	4	24	12
Contract activation latency, days	7 to 60	2	26	12
Capital per node, USD	15M to 100M	0	28	12
Fixed quality and labour, USD/site-yr	1.5M to 6M	0	28	12

own. Five norepinephrine cells lose feasibility below the demand they were designed for, and on sodium bicarbonate fourteen of twenty do. These are sized designs rather than resilient ones, sitting near a capacity cliff that the choice between them moves by a few per cent of demand.

Two of the seven sensitivity axes the specification names are covered only indirectly, which is stated rather than papered over. Uptime and yield enter the model only through saleable capacity in Eq. 1, so a proportional change in either is the same lever as the same proportional change in batches per site-year, the row that already moves 39 of 40 cells; neither was bisected separately in the threshold grid. Routine demand volatility likewise was not bisected: it appears in the ablation below as the demand-shock and correlated-regional-shock mechanisms, both of which rank low, while the demand *level* was swept and dominates. The other five axes, capacity utilization, release time, quality and fixed cost, and raw-material lead time, are swept directly in Table 9 or ablated in Table 10.

Table 10: Leave-one-out attribution over 40 cells, 60 paired runs, from `abl_post_R009/attribution.csv`. Positive deltas mean removing the mechanism improves service; cost is the mean change in annual total cost, thousands of USD. Five mechanisms with mean Δ fill below 0.0003 are omitted, and the three cost mechanisms, which move every service metric by exactly zero, are reported as one row.

Mechanism removed	Family	Mean Δ fill	Max Δ fill	Mean ΔP	Max ΔP	Short. d/yr	Cost kUSD
Capacity shortfall	capacity	+0.01128	+0.11066	+0.150	+0.967	-10.2	+379
Raw-material buffer	supply	+0.00535	+0.03508	+0.049	+0.167	-4.2	+535
Common-cause events	dependence	+0.00522	+0.04325	+0.052	+0.233	-4.6	+57
Supplier disruption	supply	+0.00393	+0.02137	+0.031	+0.083	-2.9	+33
Idiosyncratic site failure	capacity	+0.00371	+0.02342	+0.041	+0.217	-3.0	+41
Backorder window, 7 to 30 d	structure	+0.00312	+0.01293	+0.039	+0.117	+2.1	+4
Deviations, rejections, yield var.	quality	+0.00148	+0.00841	+0.015	+0.100	-1.3	-13
Regional review rule	inventory	+0.00143	+0.00887	+0.032	+0.233	-3.5	+141
Demand shocks	demand	+0.00127	+0.00860	+0.004	+0.083	-1.1	-9
Sterility incubation, 14 days	release	+0.00099	+0.00323	+0.013	+0.067	-1.4	+17
Correlated regional shocks	demand	+0.00084	+0.00493	+0.004	+0.033	-0.7	-4
Commissioning delay	time	+0.00054	+0.00495	+0.015	+0.150	-0.8	+285
503B availability	regulatory	+0.00034	+0.01200	+0.004	+0.133	-0.3	+5
Chemical release queue, 2 days	release	+0.00031	+0.00149	+0.000	+0.033	-0.5	+6
The three cost mechanisms	cost	0.00000	0.00000	0.000	0.000	0.0	-5431
Component lead time	supply	-0.00101	+0.00053	-0.005	+0.017	+0.6	-15
Ten-year horizon	structure	-0.01504	+0.00276	-0.051	+0.083	+12.7	+375

Capacity shortfall is more than twice the next mechanism on mean fill and three times it on the tail

probability, and its largest single-cell effect, +0.111 fill and +0.967 on $P(\text{meet})$ at sodium bicarbonate S19, is an order of magnitude larger than any other mechanism’s largest. Two rows are warnings rather than findings. Component lead time carries a negative sign, meaning a shorter lead slightly worsens service; revision R011 fixed the policy half of that defect and a residual remains, because under a forced outage a deferred order lands when the supplier resumes whatever its nominal lead, so a longer lead carries a larger protective pipeline. That is a property of the disruption model and is recorded rather than argued away. The ten-year horizon row is a measurement convention, not a mechanism.

Table 11: Interaction of the three deciding mechanisms, 60 paired runs, averaged over 40 cells, from `abl_post_R009_pairs/`. The additive prediction adds the single-mechanism deltas to the base; the last two columns are observed minus predicted.

Mechanisms removed	Mean fill	$P(\text{meet})$	Short. d/yr	Annual USD M	Additive fill	Fill inter.	P inter.
None (base)	0.98666	0.8058	13.60	19.48	.	.	.
Capacity adequacy only	0.99794	0.9558	3.37	19.86	0.99794	0.00000	0.0000
Regional review rule only	0.98809	0.8379	10.10	19.62	0.98809	0.00000	0.0000
Commissioning only	0.98720	0.8208	12.80	19.76	0.98720	0.00000	0.0000
Capacity plus review rule	0.99906	0.9796	0.69	20.03	0.99937	-0.00031	-0.0083
Capacity plus commissioning	0.99811	0.9617	3.10	20.14	0.99847	-0.00036	-0.0092
Review rule plus commissioning	0.98842	0.8467	9.80	19.91	0.98863	-0.00021	-0.0062
All three	0.99913	0.9817	0.62	20.33	0.99990	-0.00078	-0.0212

Ranking mechanisms one at a time does not show whether they substitute. Removing all three lifts the average tail probability from 0.806 to 0.982 and cuts average shortage days from 13.6 to 0.6, at an average cost increase of 0.85 million USD per year. The interactions are all negative and all small, so the three substitute weakly rather than reinforce: fixing capacity first removes most of the value of fixing the other two, and capacity alone recovers 0.0113 of the 0.0125 fill and 0.150 of the 0.176 tail probability the full triple delivers. That ordering is the practical content of the sensitivity analysis, and it exposes a gap: the raw-material buffer ranks second in Table 10 and is a design variable in every search space, but it was left out of the interaction study, so its substitution behaviour with capacity is unmeasured.

Table 12: Decision-reversal map, from `ds_post_R009/reversal.json`, 40 paired runs per cell. Each entry is the preferred strategy with the number of feasible strategies of 20 in brackets; demand is a multiple of the base for that product. Entries were identical at all three levels of fixed quality and labour cost per site-year (1.5M, 3M and 6M USD), so that axis is collapsed.

Product and demand	Common-cause 0.02	0.10	0.30
Norepinephrine, 0.56	S15 (19)	S15 (19)	S15 (18)
Norepinephrine, 1.00	S15 (20)	S15 (20)	S17 (1)
Norepinephrine, 2.00	none (0)	none (0)	none (0)
Bicarbonate, 0.50	S0 (20)	S15 (20)	S15 (19)
Bicarbonate, 1.00	S11 (15)	S11 (15)	none (0)
Bicarbonate, 2.00	none (0)	none (0)	none (0)

Table 12 contains three results. A four-fold change in the fixed quality and labour cost of a site reverses no decision anywhere: in a study about whether to build manufacturing capacity, the cost of running a quality unit does not decide the answer and the amount of capacity does. At twice the design demand nothing in the battery is feasible on either product, which is the demand cliff seen from the other side. And dependence between sites is the one hazard that can collapse the feasible set on its own: at base demand and a common-cause rate of 0.30 per year, norepinephrine drops from twenty feasible strategies to one and sodium bicarbonate from fifteen to none. Correlated failure rather than site count is what an architecture has to defeat, and adding sites that still share an API source, a container supplier, a geography or a quality unit does not defeat it. Most rows prefer S15, which builds no new plant, because it is the cheapest design meeting the requirement, which is the frozen decision rule working as specified. Two qualifications stop that being a

recommendation: the map runs at 40 paired runs, where the standard error is 0.047, so its feasible sets are coarse; and S15 is a product-selection construct, a presentation whose capacity is adequate and whose buffer is payable, so its position at the top says which product to pick and not which network to build.

On release time specifically, a chemical release-assurance layer can only reduce the assay component of the sixteen-day path, and the engine's ablation for it zeroes assay, endotoxin, environmental monitoring and quality review together, removing 2 of 16 days for +0.00031 mean fill and +0.00149 in the single cell where it does most. Removing the entire 14-day sterility incubation, which no release-assurance layer can do and which only a change of sterilization route with parametric release could reach, moves mean fill by +0.00099 and at most +0.00323. A perfect quality system with no deviations, rejections or yield variance moves it by +0.00148 and at most +0.00841, the largest figure this package has recorded for any release-side or quality-side ablation in any run vintage. On mean fill, capacity is worth eight times the quality ablation and eleven times the sterility one, and the one-way analysis agrees: the sterility component changes feasibility in 6 of 40 cells against 40 of 40 for demand. In the dominance analysis S6 is dominated by S5 on both products. Defect MD-18 means part of that identity is an artifact, but no run in any vintage has shown the release layer paying for itself.

Four limits on this analysis are load-bearing and none is closed. The inputs are illustrative, so it says what the model is sensitive to and not where the real world sits inside those ranges; the most consequential unknown is the one the analysis calls decisive, the true capacity headroom of the incumbent networks for these two presentations, and the package holds no evidence for it above tier 5. The resolution is coarse and inconsistent, at 20, 40, 60 and 100 paired runs with standard errors of 0.067, 0.047, 0.039 and 0.030, and fifteen of forty cells change classification between the two most recent optimizations, so individual cell verdicts are provisional throughout. Several model defects bearing on these numbers remain open and are listed in Table 14. And nothing here has been reviewed by anyone.

6 Regulatory pathways: what each permits, restricts, and leaves uncertain

This is a preliminary regulatory analysis. It is not legal advice. No statement in it has been reviewed by a qualified generic-drug, CMC, sterile-manufacturing or 503B professional. All sixteen gates carry status: UNCERTAIN with a null reviewer and a null review date. UNCERTAIN is never PASS, so `regulatory.evaluate_all` returns NO_CONCLUSION for every strategy in every run quoted in this memo. Nothing below asserts that any architecture is permitted, compliant, clinically deployable, or ready to manufacture a drug. Gates are hard constraints: none is relaxed anywhere to improve an economic result, and the engine carries no penalty-cost substitute for one.

6.1 Approved-generic contract manufacturing

Permits. Supply of an approved presentation requires an approved application and a site named in it. The decision map (`docs/regulatory/approved_generic_cmo_map.md`) records four positions: hold an ANDA or 505(b)(2) NDA; be added as a contract fill-finish site to a holder's application by supplement; take transfer of an application; or supply components or bulk only. Owners are identifiable from the frozen 2026-09-01 Drugs@FDA and Orange Book snapshots, where the two modelled presentations sit behind 26 injectable applications from 23 sponsors (norepinephrine) and 19 from 13 (sodium bicarbonate). Which of them would contract, license or compete is an unresolved commercial question this package does not answer.

Restricts. Sites must be registered and drugs listed under FD&C Act 510 and 21 CFR Part 207, and site additions are governed by 21 CFR 314.70. Batch disposition is non-delegable under 21 CFR 211.22(a), an extramural facility is an extension of the manufacturer under 21 CFR 200.10(b), and a quality agreement cannot delegate CGMP responsibility. Nominally separate sites under one application therefore share one quality unit and share its hazard, which revision R007 had to correct in five configurations that had asserted an independence their own regulatory-owner field denied.

Leaves uncertain. The reporting category for adding a third-party sterile fill-finish site is expert interpretation until a reviewer assigns it. No framework grants automatic statistical interchangeability across sites, so gate G14 blocks any fleet-level equivalence claim while UNCERTAIN, removing the pooled-validation and shared-quality-unit savings that are the whole economic case of the modular fleet family. The distributed-manufacturing registration instrument is only a proposed rule (FR doc 2026-14073, 91 FR 42888), the base case is forbidden to depend on it, and a word search of its text returns no occurrence of sterile, aseptic, 503B, outsourcing facilit, or real-time release.

What changes in the model. G01 excludes a strategy outright if the relationship is absent, so every non-503B result in this memo presumes an application relationship that does not exist. G02, G07, G13 and G16 are lead-time gates the engine serves with tier-5 stand-ins rather than with a category read from a regulatory assessment. G03 to G05 set the earliest commercial start, modelled as a one-time cost plus a delay and not as a probability of failure. G06 caps shelf life, which binds for several architectures in the current threshold grid. G16 blocks an alternate API or container source until its lots are qualified under 21 CFR 211.84 and 211.94 and a Type II drug master file letter of authorisation issues; it applies to S9, S18 and S19, and S9 is one of only two architectures meeting the service target on both products, so its source independence rests on an UNCERTAIN gate.

6.2 503B outsourcing

Permits, and for exactly how long. A registered outsourcing facility may compound from a bulk substance only if that substance is on the 503B bulks list or the drug is on FDA's shortage list, and 503B(a)(5) bars copying an approved product except through the shortage exception. The predicate must hold at compounding, at distribution, and at dispensing. **Eligibility is therefore tied to the shortage list or the bulks list, and it evaporates when a shortage resolves.** When the listing lapses, compounding stops, and inventory already compounded may not be distributed or dispensed under the exception after that date as the statute is written; whether transition relief exists is an unresolved question for a reviewer. The package models 503B as a time-varying legal state, never a durable pathway and never a bridge to an approved application.

The frozen record. On the 2026-09-02 snapshots, sodium bicarbonate 8.4 percent carries 19 shortage rows of which 15 are current, first posted 2017-03-01, and is explicitly NOT INCLUDED on the final bulks list (88 FR 20531), so the shortage listing is its only predicate. Norepinephrine 1 mg/mL carries zero shortage rows and appears on neither bulks table, so the pathway is unavailable for it today. Furosemide 10 mg/mL, a reserve candidate, carries 33 rows of which 30 are current.

What changes in the model. The engine computes `eligible_today = on_shortage_list AND static_gates_pass AND state_permission` and re-checks the listing on each compounding, shipment and issue day. The only 503B cells ever feasible in this study were produced by the forbidden world in which the product is listed at time zero and the resolution rate is set to zero, that is, by assuming permanence. G12 removes regions without state licensure from the service set, but no state-by-state map is frozen here, so that restriction is unmodelled rather than modelled as absent. The beyond-use date is also not a scalar in the source: 6 days for aseptic product released without a completed sterility test, 14 for terminally sterilized product on a validated indicator-monitored cycle, 28 for either route with a completed passing test. That instrument is revised draft guidance stamped nonbinding and not for implementation, so it is not in-force policy, and the model holds `bud_503b_days` constant where the source is a lookup on route by test completion by storage condition.

6.3 What neither pathway resolves

G15 governs any release-assurance component, its treatment is fallback to conventional release, and the R3 scenario stays disabled unless it passes, so no strategy here takes credit for a model-assisted release decision. That constraint costs almost nothing, which is itself a result: on the illustrative release inputs, removing the entire chemical release queue removes two of the sixteen base release days and G15 in full removes zero,

Table 13: The sixteen gates, their treatment in the model, and their legal-status label. Source: `config/regulatory_gates.yaml`. Every status is UNCERTAIN with a null reviewer and a null review date.

Gate	Pathway	Treatment	Legal basis type
G01 application or CMO relationship	approved	exclude if fail	law or regulation
G02 site in application, registered	approved	lead time	law or regulation
G03 process and cleaning validation	approved	hard feasibility	final guidance
G04 aseptic or terminal validated	approved	hard feasibility	final guidance
G05 analytical methods transferred	approved	hard feasibility	law or regulation
G06 stability and container closure	approved	cap	law or regulation
G07 change category and submission	approved	lead time	law or regulation
G16 alternate source qualified	approved	lead time	law or regulation
G08 outsourcing facility registered	503B	exclude if fail	law or regulation
G09 bulk eligible at all three dates	503B	dynamic state	law or regulation
G10 not barred as essentially a copy	503B	dynamic state	final guidance
G11 stability, BUD and sterility	503B	cap	law or regulation
G12 state licensure and distribution	503B	region restriction	law or regulation
G13 per-unit registration supportable	distributed	lead time	proposed rule
G14 cross-site comparability accepted	distributed	no fleet claim	unresolved question
G15 release model validated	release	fallback conventional	expert interpretation

because the one component it may reduce sits underneath two longer ones while sterility incubation is the critical path.

Neither pathway gives the study’s subject standing to change a sterilization route. Route conversion is a prior approval supplement under 21 CFR 314.70(b)(2)(iii), with 314.97(a) applying the same regime to ANDAs, and parametric release is a second prior approval supplement. Both are filed by an application holder, and a 503B facility can hold neither. That matters because route conversion with parametric release is the only intervention found across fourteen architecture families that removes the binding release constraint, and it is unavailable to a party holding no application.

7 Limitations

7.1 Inputs, and the absence of any human evidence

Eighty-seven of ninety-one registered parameters carry evidence tier 5, illustrative, and zero are missing. That covers every demand, cost, capacity, yield, uptime, lead-time, price, shelf-life and disruption input; the only sourced numeric inputs driving the runs are four release-time components. No absolute cost, break-even price, capital requirement, fill rate, shortage-days-avoided figure or incremental cost per shortage-day avoided in this memo is a statement about sterile-injectable manufacturing.

Zero interviews have been completed against a protocol target of 25 to 30, zero of three required independent reviews, zero of sixteen gate reviews. Every counterparty named anywhere is a published third party cited by source id: nobody is a partner, no conversation is a letter of intent, no price has been quoted by a purchaser, no regulatory opinion has been obtained. All eleven protocol revisions carry `approved_by: pending founder sign-off`, and since the verdict is stated against mean fill at or above 0.99 in at least 0.90 of runs, and the feasible set changes at 0.98, an unsigned threshold is not a formality. Nine of twenty-two definition-of-finished tests pass, and eleven of the thirteen incomplete ones cannot be closed by any amount of code. The modelling has run ahead of the evidence.

Two input artifacts limit what any product comparison can mean. The two presentations differ mainly through one derived tier-5 number, status-quo utilization 1.245 against 0.778, and the package’s audit records the dossiers as near-clones on several axes, so most apparent product discrimination is an artifact of one assumption. The second product also failed the study’s own beachhead verification on 2026-07-16, so it is a control rather than a second market case.

7.2 Three retracted headline claims

This is kept rather than tidied away, because it bounds how much confidence any current number deserves. Revisions R009 to R011 fixed six defects that carried the architecture ranking: free opening inventory scaled to each design's own stock policy (NEW-1), unmatched search spaces between the frozen comparators and the new architectures (MD-3, MD-12), an infeasible tie-break with no Monte Carlo tolerance that had bought a 9.26 million USD per year cost increase for 0.00004 of fill (MD-11), a material buffer that scaled with the lead time (MD-1), an inert allocation policy (NEW-2), and readiness and take-or-pay instruments with no service channel (MD-23, MD-24). Three earlier conclusions were then withdrawn. That owned distributed nodes meet the target nowhere on either product: false, they meet it for norepinephrine. That no frozen comparator meets it: false, dual sourcing meets it for norepinephrine at 0.960. That bright-stock postponement is the strongest shape on both products: withdrawn, because once opening inventory is charged the design falls to 0.630 on sodium bicarbonate, having carried roughly 1.2 million free units, about five years of the structural capacity gap, inside a five-year window. The last three headline claims this study made did not survive its own next round of fixes.

7.3 Open model defects and structural limits

Twenty-nine defects are registered in docs/design_space/bottleneck_decomposition.md section 7 and docs/design_space/novel_architectures.md section 5. Twenty-one are recorded as fixed under revisions R003 to R011, one (MD-1) with a residual the register states rather than argues away. Eight remain open (Table 14); MD-15 and NEW-3 could change the ranking rather than only the numbers. One inconsistency is carried rather than smoothed: MD-17 is recorded as fixed by R008 in protocol/revisions.csv and also appears among the deferred items in the later status table.

Three further engine limits carry no id: material stock has no shelf life, retest period or hold limit, so upstream reserves escape the obsolescence that emptied every real one; per-site commissioning, activation lead, exercise cadence and activation failure probability are configuration constants no design variable can reach; and the common-cause group type cannot express a key starting material tier, a regulatory-action group, or a shared utility. The engine also runs one presentation through one network, so every portfolio, postponement and pooling statement here is an accounting proxy, and the pooling threshold the study computed, on the order of five comparable presentations per node, is untestable inside its own engine. Regions are identical by default: R011 added the plumbing for differing criticality weights and minimum guarantees and measured that, with identical regions, three of the four allocation policies return the same fill to nine decimal places.

A later and deeper optimization exists at 60 final runs (opt_20260906T043357Z), with the design-space and ablation batteries re-run on its designs, but those artifacts have not been re-evaluated at 100 runs and no narrative document interprets them. They disagree with the verified set: there 29 of 40 cells return optimal rather than 14, and the status quo itself clears the target on norepinephrine at 12.91 million USD per year, the cheapest cell in the run. The documentation lags the artifacts, and the feasible boundary is not sharp at any resolution the package has run.

7.4 Evidence limits outside the engine

The sterilization-route data is redacted, not merely unpublished. That distinction decides whether more searching would help, and it does not. Across 65 manufacturer labels behind 50 applications for the six longlist candidates, zero state a sterilization method, and for the two modelled presentations the screen determined nothing at any confidence: 0 of 24 labels and 0 of 18 applications. FDA writes the route down and withholds the word. The published NDA 214313 review reads “The proposed drug product is (b) (4) sterilized”, and the Product Quality Review template’s “Method of Sterilization:” field is withheld behind a (b)(4) redaction in the published NDA 207987 review. A Freedom of Information Act request returns the same redaction, because it is the holder’s trade-secret claim and not an agency backlog.

Table 14: The eight open model defects and what each could still move.

Id	Defect	What it could still move
MD-1 residual	Under a forced outage a shorter lead can still show marginally more stockouts, because a deferred order lands when the supplier resumes	Every raw-material lead-time condition in this study, in either direction
MD-5	The fill metric forgives anything delivered inside a 7-day backorder window	Fill by up to 0.016 in an unstable direction, and so any verdict near the requirement
MD-8	Node capacity is quadratic in the capacity factor, since batch size and batches per year both scale	Any future node-scale result; a node at scale 1.0 would be 1.82 times the central plant
MD-15	One product per network, and equal regional shares	The largest unquantified distortion in the cost comparison: 94.5 percent of one distributed-node cell's cost is borne by one presentation at 0.203 utilization
MD-18	The frozen release-time design variable is read only by the deterministic screen, never by the stochastic engine	The release-layer variant returns fill identical to the plain node design while costing 1.0 and 1.7 million USD per year more
MD-22	The screen's common-impact factor is derived from topology for new architectures and hand-set for frozen comparators	No deterministic-screen common-impact comparison across the two arms is valid
MD-25	Shared operating-layer cost is charged only outside release scenario R0, and every design-space strategy is R0	A declared shared operating layer carries hazard with no cost line and no benefit
NEW-3	No readiness-decay hazard, so exercising a line buys production volume and not reliability	The whole readiness case for the rotating and standby designs, including the architecture ranked first below
NEW-4	Every declared common-cause group draws its own Poisson stream at one global rate, so group count is not neutral	A design declaring an honest correlation is penalised for declaring it, distorting comparison between differently documented topologies

No price evidence, weak backcasts, a methods-proof release layer. A landscape review over 313 sources found no published rate card, minimum order value, reservation fee or per-unit price for US sterile fill-finish, 503B product, enterprise quality systems, shortage-data platforms or modular hardware, so every unit price and break-even figure here is an assumption. Reimbursement, market price, willingness to pay and production cost are kept as distinct concepts; CMS average sales price and Part B spending appear only as labelled utilization proxies, and no reimbursement figure in this memo is a manufacturing cost. The backcasts compare three ongoing shortages of one class against a requirement of four episode classes, and no documented single-site failure episode has been obtained. The release-assurance evidence rests on 615 of the 655 tablets in a 2002 public near-infrared benchmark, with no real lots, no real drift, no real operators and not the target chemistry; it says nothing about a sterile injectable, and abstention is never release.

Known defects in the study's own claims, and authorship. Fourteen claim-register rows are recorded as unsupported, contradicted or retracted, one of them a citation error inside the regulatory material: the repository's release-constraints memo attributes to FDA's 2004 aseptic processing guidance a sentence that is not in it. The conclusion below does not rest on that sentence; the citation is still wrong. Every artifact in this package, including the code, the configurations, the regulatory decision maps and this prose, was produced with an AI coding agent, and the disclosure record marks essentially all of it as not yet reviewed by the author. The legal-basis text in the gate configuration is that agent's classification and is unverified, and a planned adversarial verification pass over the audit findings did not complete.

8 Conclusion

Stated once more so no sentence below is mistaken for a claim about the world: every number here is engine behaviour under evidence tier 5 illustrative inputs, all sixteen gates are UNCERTAIN and unreviewed, and no interview or independent review has taken place.

8.1 Where regional manufacturing does not beat the alternatives

Three results survive every revision of the engine, including the ones that overturned its other conclusions.

A network whose saleable capacity is below demand cannot be stocked out of the deficit. On the capacity-short product a full year of safety stock with daily base-stock review leaves the status quo at mean fill 0.869 with a tail probability of zero, and on the current reversal grid the feasible set is empty in all nine high-demand cells of each product, so at twice base demand no strategy of the twenty meets the target anywhere. That region argues for capacity, not for regional capacity.

Regional nodes lose the days before they exist, and their conditions are tight. On the capacity-short product the distributed node design needs commissioning at or below 741 days, a supplier disruption rate at or below 0.127 against an assumed base of 0.200, shelf life at or above 17.6 months and changeover at or below 4.5 days, so it sits outside its own feasible region on the supplier axis at the base assumption. Its cost there is the highest in the battery, 32.0 million USD per year against 19.4 million for the cheapest cell that meets the target on that product.

Adding a validated release layer to those nodes buys nothing. The release-layer variant returns mean fill identical to the plain node design to four decimals on both products while costing 1.0 and 1.7 million USD per year more, for two reasons, a model defect (MD-18) and a regulatory constraint (G15 held UNCERTAIN), neither of which is an argument for the layer.

8.2 Where it clears the bar, and the two architectures that clear both products

Regional manufacturing is not eliminated. On the product that is not capacity-short, the distributed node design meets the frozen target at a tail probability of 0.940, which is not borderline, at 20.0 million USD per year. What had previously made it fail was in part an unequal search space: it had been denied the inventory freedom its competitors were given, and that is the clearest case in the package of a conclusion that was an artifact of the harness rather than a statement about distribution. The honest form is narrow. Regional capacity can meet the target where the incumbent plant is not capacity-short, at about 9 percent above rotating contracted campaigns, 11 percent above plain dual sourcing and 32 percent above the cheapest feasible cell; it cannot meet it at competitive cost where the plant is capacity-short, which is the case shortages actually present.

Of forty cells re-evaluated at 100 paired runs, fourteen meet the frozen target and only two architectures meet it on *both* products: rotating prequalified campaigns across three already-registered third-party fill lines (S17), and dual finished-dose sourcing with an explicit key starting material tier (S9). Neither is a plant the sponsor builds.

Two features of Table 15 matter more than the ranking. Three of the four demand ceilings sit at or below the base demand assumption, so the conditions and the base verdicts disagree at the resolution of the grid. And for S9 on the resolved product, nine of twelve swept axes return infeasible at both ends of their recorded range while the same cell is feasible at base at 100 paired runs, which the package does not resolve. Read the conditions as an indication of which axes bind, not as numbers to plan against. Neither architecture is contractable today: no purchaser has been asked and no price has been quoted.

One condition applies to every architecture at once, and it is the strongest single result in the current reversal grid. At base demand, raising the multi-site common-cause rate from its base of 0.1 to the top of its recorded range of 0.3 events per group-year collapses the feasible set from 20 strategies to 1 on norepinephrine and from 15 to 0 on sodium bicarbonate, and the one survivor is the rotating campaign network. Against

Table 15: The two architectures meeting the service target on both products. Results at 100 paired runs from `results/design_space/reeval_post_R009.json`; conditions at 20 paired runs per evaluation from `ds_post_R009/thresholds.csv`. All inputs tier 5 illustrative. Base demand 900,000 units per year (norepinephrine) and 1,200,000 (sodium bicarbonate); base common-cause rate 0.1 per year; base 45 batches per site-year; base changeover 3 days.

Architecture	Product	Cost M USD/yr	Fill mean	P(meet)	Binding conditions inside the swept ranges
S17 rotating cam-paigns	norepine.	18.4	0.9975	1.000	demand at or below 926,600 per year; changeover at or below 8.45 days
S17 rotating cam-paigns	sodium bi-carb.	19.4	0.9960	0.900	demand at or below 1,191,000, below the base; common cause at or below 0.182; batches at or above 44.4
S9 dual source, KSM tier	norepine.	23.3	0.9975	0.920	demand at or below 642,200, well below the base; common cause at or below 0.068; batches at or above 51.9; the other nine axes infeasible at both ends
S9 dual source, KSM tier	sodium bi-carb.	24.6	0.9987	0.940	demand at or below 1,303,000; shelf life at or above 12.4 months; API lead at or below 125 days; common cause at or below 0.182; supplier rate at or below 0.385

that, fixed quality and labour cost per node changes the preferred strategy in zero of the eighteen demand-by-common-cause cells across both products, and neither it nor node capital crosses a feasibility boundary in any of their 40 threshold rows. Correlated failure decides this map. Fixed quality cost does not.

8.3 What would overturn this

A demand denominator. True utilization and installed capacity for the exact presentation produces 40 of the 202 crossings on the current threshold grid, more than any other swept input, and removes an architecture outright if it lands above that architecture’s demand ceiling. Every ceiling in Table 15 sits within a few percent of the base or below it.

A price and a committed volume from a purchaser with authority. On the pre-correction feasible set of thirteen cells, five moved from commercially possible to impossible as the assumed price fell from each design’s own break-even to 18.00 USD per unit, which is an assumption and not evidence. An answer of “nobody funds standing availability” leaves every capacity-carrying design unfunded by construction.

The G07 reporting category in writing. If adding a third-party sterile fill site returns as a prior approval supplement with a preapproval inspection, the fastest leg of the contracted family lengthens, and the contracted family is where both surviving architectures sit.

Closing the open defects and re-running. Until NEW-3 gives readiness a decay hazard, exercising a line buys production volume rather than reliability, and the architecture ranked first above is routine contracted supply wearing a standby label. Until MD-15 allows more than one product per network, every pooling statement is an accounting proxy. The study has already shown twice that closing defects of this size changes which architecture wins.

8.4 The conditional answer, stated plainly

On this evidence, regional manufacturing beats safety stock, dual sourcing and additional centralized capacity only under a narrow conjunction: the presentation’s incumbent capacity is not short of demand, demand sits below roughly the base assumption, correlated failure stays near the low end of its range, commissioning completes inside the measured window, and a buyer will pay a premium over contracted alternatives for something other than service, because the service is matched and the cost is not. Of the alternatives, additional safety stock alone never meets the target on the capacity-short product anywhere in the searched space, and once every comparator searches the same inventory space it is no longer distinguishable from the status

quo, returning identical cost, fill and tail probability in both current runs; dual sourcing meets the target on one product and not the other; and rotating contracted campaigns on already-registered lines meet it on both, more cheaply than nodes on either.

The defensible conclusion is therefore not an architecture. It is that the binding constraints in this model are capacity relative to demand, correlated failure across nominally separate sites and suppliers, the non-negotiable sterility hold, and a set of regulatory gates no design choice reaches, and that none of those four is improved by distributing manufacture as such. Nothing here says any architecture is FDA-compliant, clinically deployable, or ready to manufacture a drug. Whether the conclusion also holds for sterile-injectable manufacturing in the world is not something this study is entitled to say, and it will not be until the inputs are real, the gates are reviewed by qualified people, and the defects in Section 7 are closed.

A Run identifiers behind every table, and how to reproduce them

Every table in this memo is produced by a named run whose manifest is written to `results/manifests/`. A manifest records the code version, the configuration digest, the seed, the run count and the parameter overrides in force, so a table can be traced to the exact state that produced it. All runs listed here postdate protocol revisions R009, R010 and R011; every earlier run in the package is superseded for quantitative use and none appears in this memo. Commands are run from the package root with the locked environment (`make env`).

Table 16: Provenance of every numbered table. Paired runs are the Monte Carlo replications behind each cell, on common random numbers across strategies. “SE” is the binomial standard error of the tail probability at the 0.90 requirement.

Table	Run identifier or artifact	Runs	SE	Command
1 screen	screen_manifest.json	n/a	n/a	make dossiers
2 revisions	protocol/revisions.csv	n/a	n/a	version-controlled file
3 release	config/global.yaml	n/a	n/a	configuration file
4 operations	config/products/*.yaml	n/a	n/a	configuration files
5 reliability	config/global.yaml	n/a	n/a	configuration file
6 architectures	config/strategies/design_space.yaml	n/a	n/a	configuration file
?? norepinephrine	opt_20260906T043357Z	60	0.039	make optimize
?? bicarbonate	opt_20260906T043357Z	60	0.039	make optimize
8 feasible set	reeval_post_R009.json	100	0.030	scripts/reevaluate_optima.py
9 thresholds	ds_post_R009/thresholds.csv	20	0.067	make design-space
10 attribution	abl_post_R009/attribution.csv	60	0.039	make ablate
11 interaction	abl_post_R009/pairs/	60	0.039	make ablate --pairs
12 reversal	ds_post_R009/reversal.json	40	0.047	make design-space
13 gates	config/regulatory_gates.yaml	n/a	n/a	configuration file
14 defects	bottleneck_decomposition.md §7	n/a	n/a	version-controlled file
15 clears both	reeval_post_R009.json plus thresholds	100, 20	0.030	both commands above

Two cautions belong with this table. The resolutions are not uniform: a verdict from the threshold grid at 20 paired runs and a verdict from the verified feasible set at 100 are not interchangeable, and where they disagree the memo reports both rather than choosing. And the narrative documents in `docs/design_space/` were written against the `reeval_post_R009` artifact; the later optimization `opt_20260906T043357Z` and the batteries re-run on its designs are not yet interpreted by any narrative document, which is why some figures in Section 7 differ from `docs/design_space/post_R009_correction.md`. That document remains the governing statement of the verified feasible set, and it supersedes the earlier synthesis and the regions document wherever they disagree.

Reproduction requires the frozen snapshots under `data/`, which are pinned by SHA-256 hash in the source register `config/sources.yaml` and verified by `make protocol-hash`. Seeds are derived through `rng.RunStreams` from the run index, a named stream, a BLAKE2b hash of the entity identifier and an

ordinal, so a rerun at the same run count reproduces the same world for every strategy. The complete assumption register, generated from the configuration files rather than written by hand, is `ASSUMPTIONS.md` in the package root, rebuilt by `python scripts/build_assumptions_table.py`.